Title : Arithmetic Logic Unit (ALU) along with Testbench

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Aim : The ALU will have inputs for two operands (A and B), an opcode for the operation (e.g., 00 for ADD, 01 for SUB), and an output for the result with suitable testbench.

Tools : Quartus and Questasim

The 5 stage pipelined ALU:

```
module pipelined_alu_16bit(
        input clk,
        input reset,
        input [15:0] a_in,
        input [15:0] b_in,
        input c_in,
        input [3:0] opcode,
        output reg [31:0] result,
        output reg c_out
);

// IF
reg [15:0] if_id_a, if_id_b;
reg if_id_cin;
reg [3:0] if_id_op;

always @(posedge clk or posedge reset) begin
    if (reset) begin
        if_id_a   <= 0;
        if_id_b   <= 0;
        if_id_cin <= 0;
        if_id_op  <= 0;
```

```
        if_id_op  <= 0;
    end else begin
        if_id_a   <= a_in;
        if_id_b   <= b_in;
        if_id_cin <= c_in;
        if_id_op  <= opcode;
    end
end

//ID
reg [15:0] id_ex_a, id_ex_b;
reg id_ex_cin;
reg [3:0] id_ex_op;

always @(posedge clk or posedge reset) begin
    if (reset) begin
        id_ex_a <= 0; id_ex_b <= 0; id_ex_cin <= 0; id_ex_op <= 0;
    end else begin
        id_ex_a   <= if_id_a;
        id_ex_b   <= if_id_b;
        id_ex_cin <= if_id_cin;
        id_ex_op  <= if_id_op;
```

```
43          id_ex_op  <= if_id_op;
44      end
45  end
46
47  //EX
48  reg [31:0] ex_mem_res;
49  reg ex_mem_co;
50
51  always @(posedge clk or posedge reset) begin
52      if (reset) begin
53          ex_mem_res <= 0; ex_mem_co <= 0;
54      end else begin
55          case (id_ex_op)
56              4'b0000: {ex_mem_co, ex_mem_res[15:0]} <= id_ex_a + id_ex_b + id_ex_cin;
57              4'b0001: {ex_mem_co, ex_mem_res[15:0]} <= id_ex_a - id_ex_b - id_ex_cin;
58              4'b0010: begin ex_mem_res <= id_ex_a * id_ex_b; ex_mem_co <= 0; end
59              4'b0011: begin ex_mem_res <= (id_ex_b != 0) ? (id_ex_a / id_ex_b) : 32'hFFFFFFFF; ex_mem_co <= 0; end
60              4'b0100: begin ex_mem_res <= {16'b0, (id_ex_a & id_ex_b)}; ex_mem_co <= 0; end
61              4'b0101: begin ex_mem_res <= {16'b0, (id_ex_a | id_ex_b)}; ex_mem_co <= 0; end
62              4'b0110: begin ex_mem_res <= {16'b0, (id_ex_a ^ id_ex_b)}; ex_mem_co <= 0; end
63              4'b0111: begin ex_mem_res <= {16'b0, (~id_ex_a)}; ex_mem_co <= 0; end
64              default: begin ex_mem_res <= 0; ex_mem_co <= 0; end
```

```
58              4'b0010: begin ex_mem_res <= id_ex_a * id_ex_b; ex_mem_co <= 0; end
59              4'b0011: begin ex_mem_res <= (id_ex_b != 0) ? (id_ex_a / id_ex_b) : 32'hFFFFFFFF; ex_mem_co <= 0; end
60              4'b0100: begin ex_mem_res <= {16'b0, (id_ex_a & id_ex_b)}; ex_mem_co <= 0; end
61              4'b0101: begin ex_mem_res <= {16'b0, (id_ex_a | id_ex_b)}; ex_mem_co <= 0; end
62              4'b0110: begin ex_mem_res <= {16'b0, (id_ex_a ^ id_ex_b)}; ex_mem_co <= 0; end
63              4'b0111: begin ex_mem_res <= {16'b0, (~id_ex_a)}; ex_mem_co <= 0; end
64              default: begin ex_mem_res <= 0; ex_mem_co <= 0; end
65          endcase
66      end
67  end
68
69  // --- Stage 4: MEM ---
70  reg [31:0] mem_wb_res;
71  reg mem_wb_co;
72
73  always @(posedge clk or posedge reset) begin
74      if (reset) begin
75          mem_wb_res <= 0; mem_wb_co <= 0;
76      end else begin
77          mem_wb_res <= ex_mem_res;
78          mem_wb_co  <= ex_mem_co;
79      end
```

```
            4'b0010: begin ex_mem_res <= id_ex_a * id_ex_b; ex_mem_co <= 0; end
            4'b0011: begin ex_mem_res <= (id_ex_b != 0) ? (id_ex_a / id_ex_b) : 32'hFFFFFFFF; ex_mem_co <= 0; end
            4'b0100: begin ex_mem_res <= {16'b0, (id_ex_a & id_ex_b)}; ex_mem_co <= 0; end
            4'b0101: begin ex_mem_res <= {16'b0, (id_ex_a | id_ex_b)}; ex_mem_co <= 0; end
            4'b0110: begin ex_mem_res <= {16'b0, (id_ex_a ^ id_ex_b)}; ex_mem_co <= 0; end
            4'b0111: begin ex_mem_res <= {16'b0, (~id_ex_a)}; ex_mem_co <= 0; end
            default: begin ex_mem_res <= 0; ex_mem_co <= 0; end
        endcase
    end
end

// --- Stage 4: MEM ---
reg [31:0] mem_wb_res;
reg mem_wb_co;

always @(posedge clk or posedge reset) begin
    if (reset) begin
        mem_wb_res <= 0; mem_wb_co <= 0;
    end else begin
        mem_wb_res <= ex_mem_res;
        mem_wb_co  <= ex_mem_co;
    end
```

```
always @(posedge clk or posedge reset) begin
    if (reset) begin
        mem_wb_res <= 0; mem_wb_co <= 0;
    end else begin
        mem_wb_res <= ex_mem_res;
        mem_wb_co  <= ex_mem_co;
    end
end

// --- Stage 5: WB ---
always @(posedge clk) begin
    if (reset) begin
        result <= 0;
        c_out  <= 0;
    end else begin
        result <= mem_wb_res;
        c_out  <= mem_wb_co;
    end
end

endmodule
```

```
1   `timescale 1ns / 1ps
2
3   module tb_pipelined_alu;
4
5       // Inputs
6       reg clk;
7       reg reset;
8       reg [15:0] a_in;
9       reg [15:0] b_in;
10      reg c_in;
11      reg [3:0] opcode;
12
13      // Outputs
14      wire [31:0] result;
15      wire c_out;
16
17      // Instantiate the Unit Under Test (UUT)
18      pipelined_alu_16bit uut (
19          .clk(clk),
20          .reset(reset),
21          .a_in(a_in),
22          .b_in(b_in),
```

tb_pipelined_alu.v | pipelined_alu_16bit.v

```
6
7       // Instantiate the Unit Under Test (UUT)
8       pipelined_alu_16bit uut (
9           .clk(clk),
0           .reset(reset),
1           .a_in(a_in),
2           .b_in(b_in),
3           .c_in(c_in),
4           .opcode(opcode),
5           .result(result),
6           .c_out(c_out)
7       );
8
9       // Clock generation (100MHz)
0       always #5 clk = ~clk;
1
2       // Task to apply inputs
3       task drive_input(input [15:0] a, input [15:0] b, input ci, input [3:0] op);
4           begin
5               @(negedge clk);
6               a_in = a;
7               b_in = b;
```

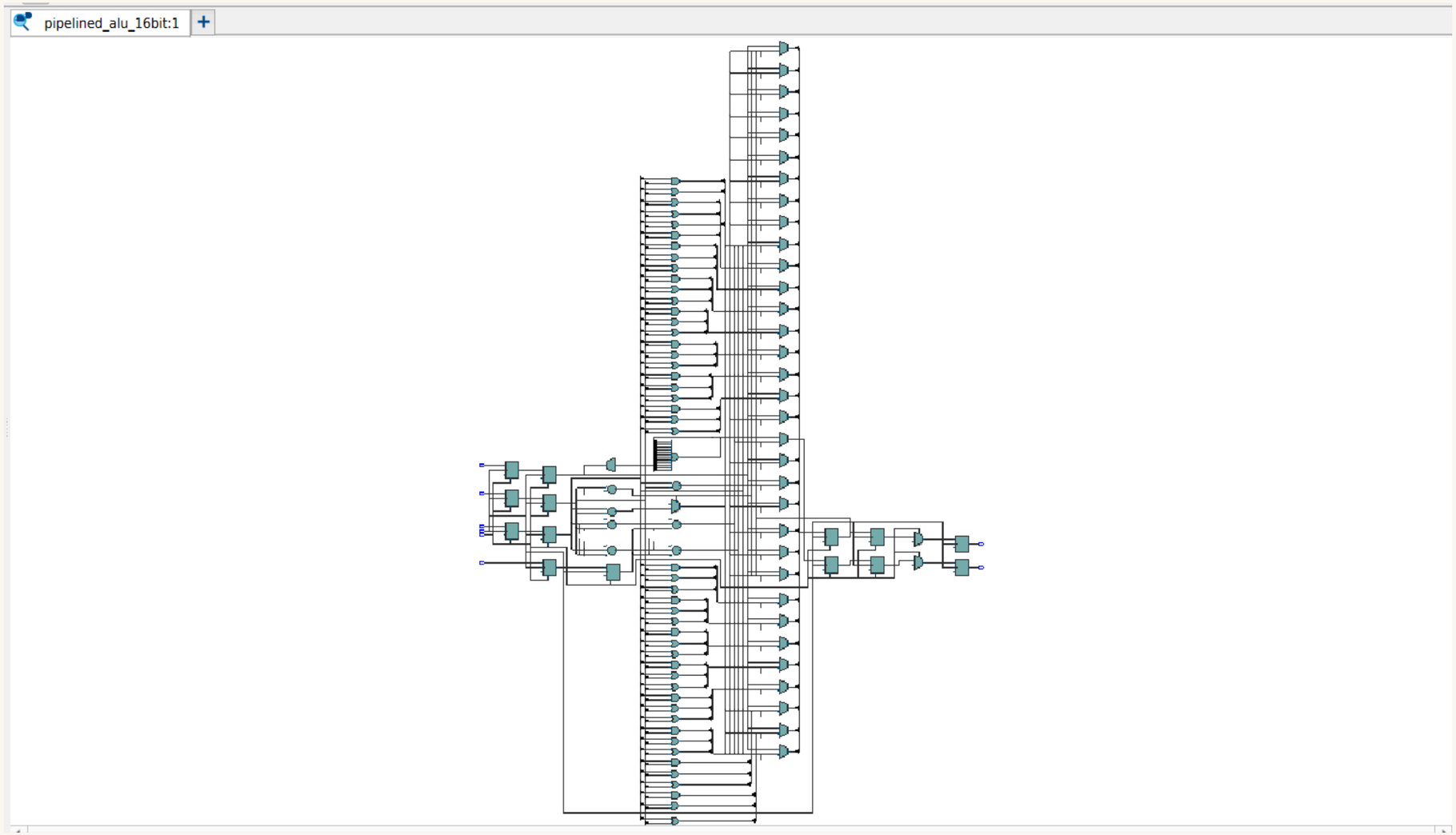
```
        begin
            @(negedge clk);
            a_in = a;
            b_in = b;
            c_in = ci;
            opcode = op;
        end
    endtask

    initial begin
        // Initialize Inputs
        clk = 0;
        reset = 1;
        a_in = 0;
        b_in = 0;
        c_in = 0;
        opcode = 0;

        // Release Reset
        #20 reset = 0;

        // --- Test Case 1: Addition (Op 0) ---
```

```
        c_in = 0;
        opcode = 0;

        // Release Reset
        #20 reset = 0;

        // --- Test Case 1: Addition (Op 0) ---
        // Input at T0, Result expected at T5
        drive_input(16'h0010, 16'h0020, 1, 4'b0000); // 16 + 32 + 1 = 49 (0x31

        // --- Test Case 2: Multiplication (Op 2) ---
        drive_input(16'h000A, 16'h0005, 0, 4'b0010); // 10 * 5 = 50 (0x32)

        // --- Test Case 3: XOR (Op 6) ---
        drive_input(16'hAAAA, 16'h5555, 0, 4'b0110); // XOR = FFFF

        // --- Test Case 4: Division (Op 3) ---
        drive_input(16'h0064, 16'h000A, 0, 4'b0011); // 100 / 10 = 10 (0xA)

        // Clear inputs and wait for pipeline to flush
        drive_input(0, 0, 0, 4'hF);

        // Wait for results to propagate through the 5 stages
```

```
        // --- Test Case 3: XOR (Op 6) ---
        drive_input(16'hAAAA, 16'h5555, 0, 4'b0110); // XOR = FFFF

        // --- Test Case 4: Division (Op 3) ---
        drive_input(16'h0064, 16'h000A, 0, 4'b0011); // 100 / 10 = 10 (0xA)

        // Clear inputs and wait for pipeline to flush
        drive_input(0, 0, 0, 4'hF);

        // Wait for results to propagate through the 5 stages
        repeat (6) @(posedge clk);

        $display("Simulation Finished. Check waveforms for 5-cycle latency.");
        $finish;
    end

    // Simple monitor to print results whenever they change
    initial begin
        $monitor("Time=%0t | Op=%b | Result=%h | Carry=%b", $time, opcode, result, c_out);
    end

endmodule
```

	Task
✔	Compile Design
✔	Analysis & Synthesis
✔	Fitter (Place & Route)
✔	Assembler (Generate programm
✔	Timing Analysis
✔	EDA Netlist Writer

Flow Status	Successful - Sat Mar 28 20:22:19 2026
Quartus Prime Version	25.1std.0 Build 1129 10/21/2025 SC Lite Edition
Revision Name	pipelined_alu_16bit
Top-level Entity Name	pipelined_alu_16bit
Family	Cyclone V
Device	5CEFA7F31I7
Timing Models	Final
Logic utilization (in ALMs)	228 / 56,480 (< 1 %)
Total registers	184
Total pins	72 / 480 (15 %)
Total virtual pins	0
Total block memory bits	0 / 7,024,640 (0 %)
Total DSP Blocks	1 / 156 (< 1 %)
Total HSSI RX PCSs	0
Total HSSI PMA RX Deserializers	0
Total HSSI TX PCSs	0
Total HSSI PMA TX Serializers	0
Total PLLs	0 / 7 (0 %)
Total DLLs	0 / 4 (0 %)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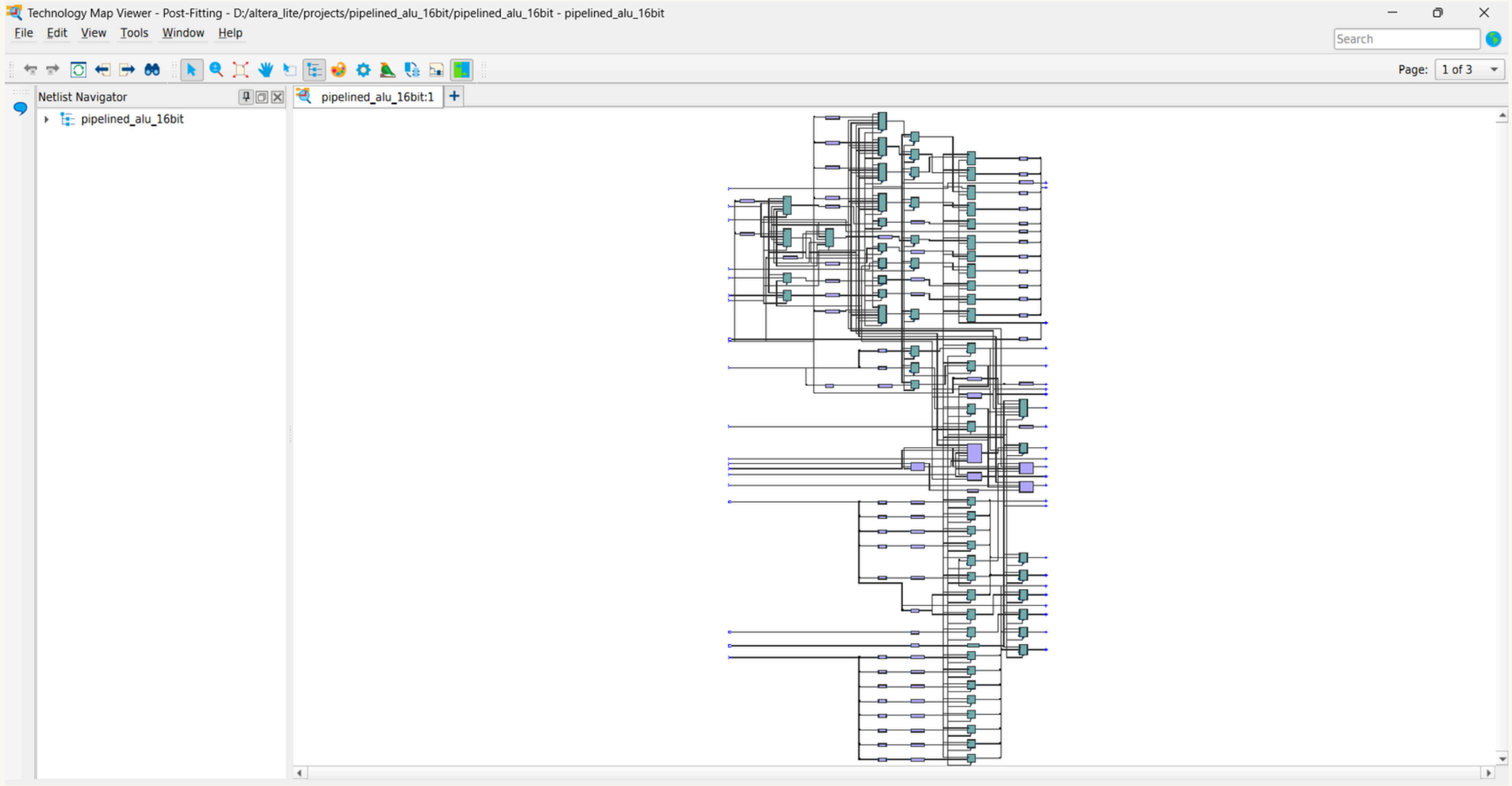
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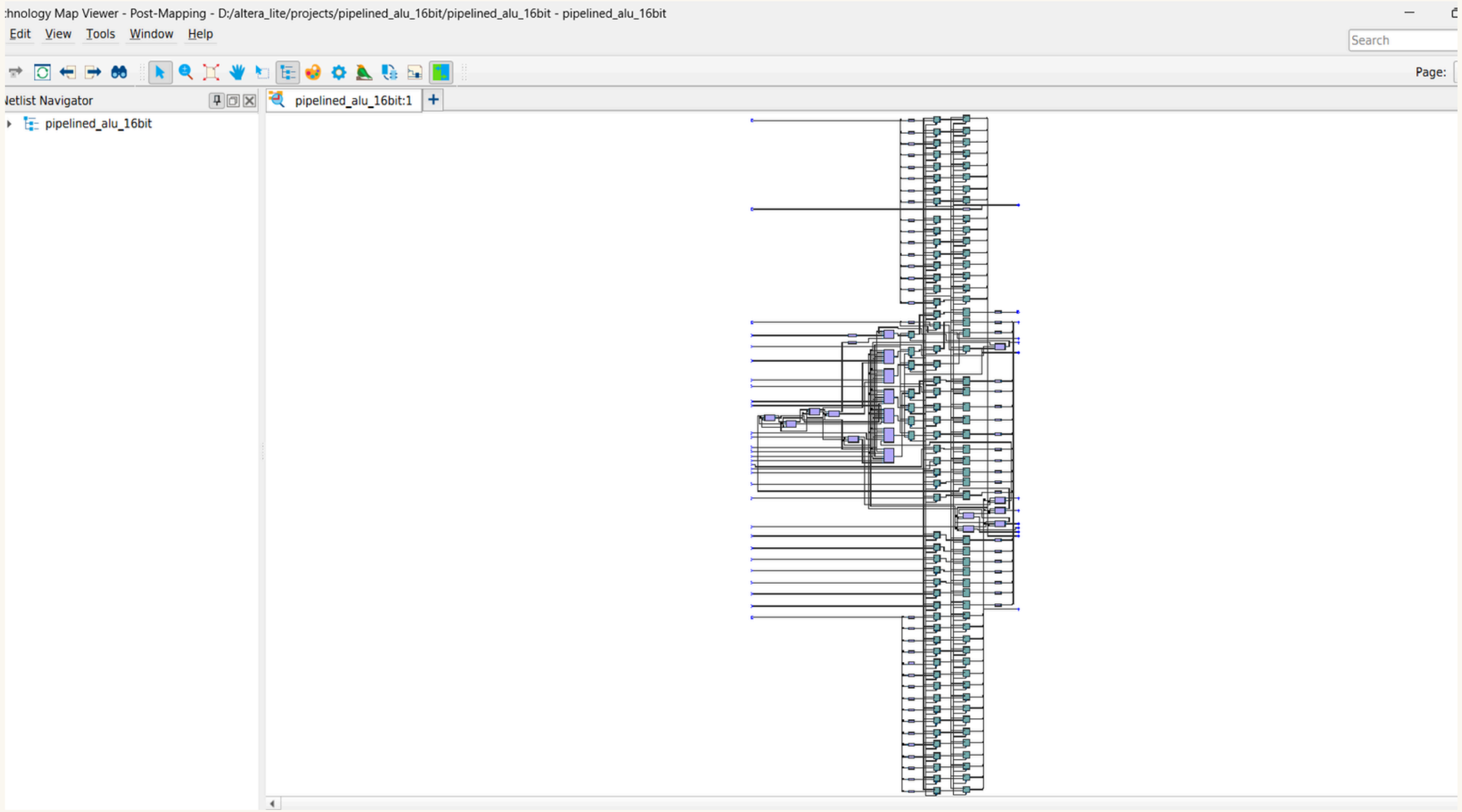
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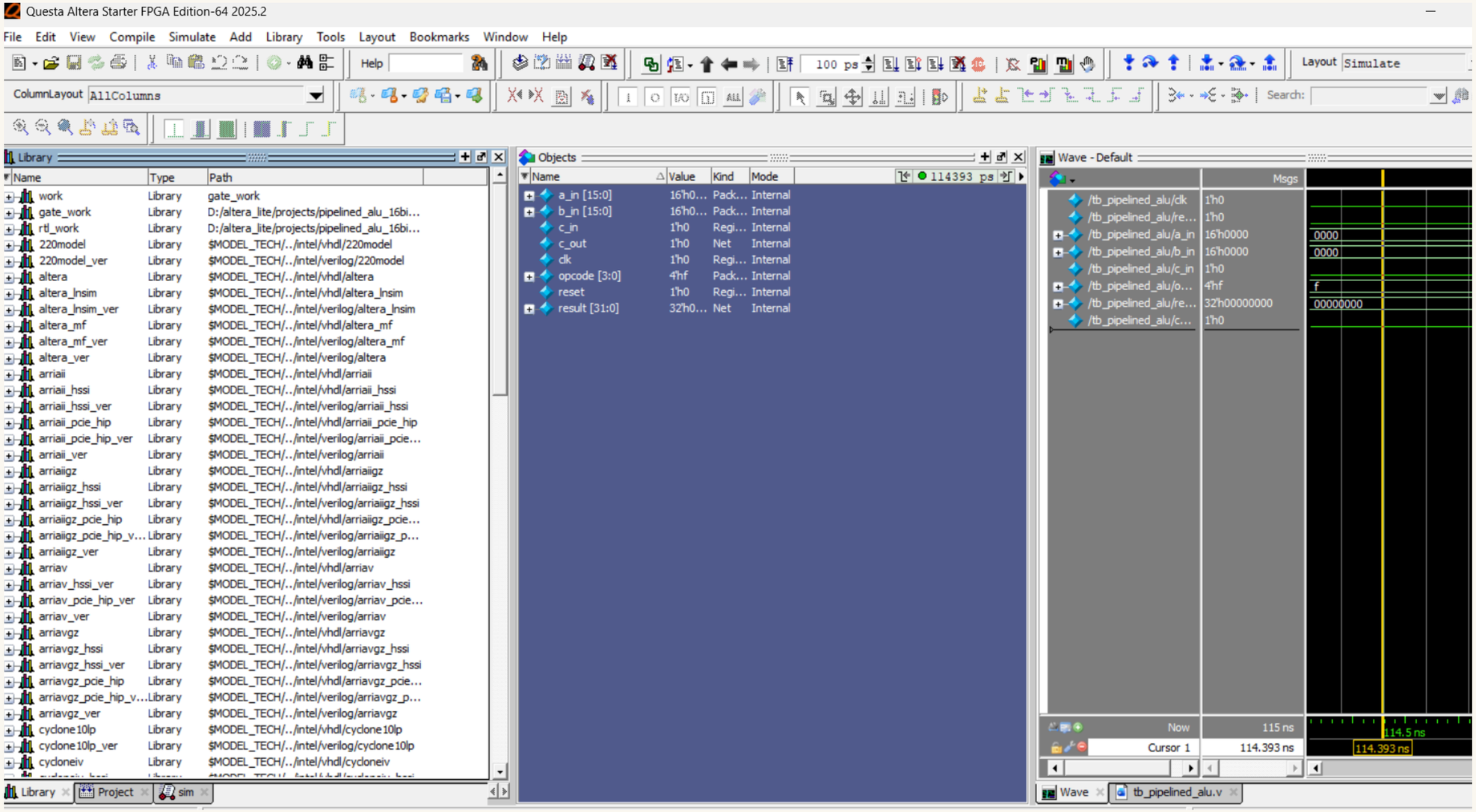
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cyclone10lp Library $MODEL_TECH/../intel/vhdl/cyclone10lp
cyclone10lp_ver Library $MODEL_TECH/../intel/verilog/cyclone10lp
cycloneiv Library $MODEL_TECH/../intel/vhdl/cycloneiv
Library Project sim
Objects
Name Value Kind Mode
114393 ps
a_in [15:0] 16'h0... Pack... Internal
b_in [15:0] 16'h0... Pack... Internal
c_in 1'h0 Regi... Internal
c_out 1'h0 Net Internal
clk 1'h0 Regi... Internal
opcode [3:0] 4'hf Pack... Internal
reset 1'h0 Regi... Internal
result [31:0] 32'h0... Net Internal
Wave - Default
Msgs
/tb_pipelined_alu/clk 1'h0
/tb_pipelined_alu/re... 1'h0
/tb_pipelined_alu/a_in 16'h0000
/tb_pipelined_alu/b_in 16'h0000
/tb_pipelined_alu/c_in 1'h0
/tb_pipelined_alu/o... 4'hf
/tb_pipelined_alu/re... 32'h00000000
/tb_pipelined_alu/c... 1'h0
0000
0000
f
00000000
Now 115 ns
Cursor 1 114.393 ns
114.5 ns
114.393 ns
Wave tb_pipelined_alu.v